



What are Pipes?

```
{{ data | pipeName:arg1:arg2 }}
```

Explanation :

1. Pipes in Angular are used to transform data in templates.
2. Think of them as formatters: they take in data, apply a transformation, and return a new value (without changing the original).

Built-in Pipes



Date Pipe



```
today = new Date();  
//HTML  
{{ today | date:'fullDate' }} <!-- e.g., Thursday, October 16, 2025 -->  
{{ today | date:'shortTime' }} <!-- e.g., 10:30 PM -->
```

Uppercase / Lowercase Pipe



```
//ts  
name = 'Shailesh';  
  
//HTML  
{{ name | uppercase }} <!-- Output: SHAILESH -->  
{{ name | lowercase }} <!-- Output: shailesh -->
```

Built-in Pipes



Currency Pipe



```
//ts
price = 1999;

//HTML
{{ price | currency:'INR' }}    <!-- Output: ₹1,999.00 -->
{{ price | currency:'USD' }}    <!-- Output: $1,999.00 -->
```

Uppercase / Lowercase Pipe



```
//ts
num = 0.756;

//HTML
{{ num | percent }}             <!-- Output: 76% -->
{{ num | number:'1.2-2' }}      <!-- Output: 0.76 -->
```

Built-in Pipes



Chaining Pipes

You can combine multiple pipes.

```
//HTML
{{ name | uppercase | slice:0:4 }} <!-- Output: SHAI -->
```

Async Pipe

Special pipe for handling Observables/Promises.

It auto-subscribes and unsubscribes (very useful with RxJS).

```
//ts
data$ = of('Hello Angular');

//HTML
{{ data$ | async }} <!-- Output: Hello Angular -->
```



Custom Pipes

When built-in pipes don't meet your needs, you can create a custom pipe.

```
//ts
import { Pipe, PipeTransform } from '@angular/core';

@Pipe({
  name: 'reverse'
})
export class ReversePipe implements PipeTransform {
  transform(value: string): string {
    return value.split('').reverse().join('');
  }
}

//HTML
{{ 'Angular' | reverse }} <!-- Output: ralugnA -->
```



Summary

- Pipes are lightweight data transformers.
- Built-in pipes handle common tasks (date, currency, uppercase, etc.).
- Custom pipes allow app-specific formatting.
- Async pipe is great for handling streams without manual subscriptions.